Predicted Future Latents as Subgoals: Hierarchical Policies Outperform Flat Policies Beyond the Training Goal Distance

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Abstract

Robot foundation models and latent world models both separate long-horizon lookahead from short action generation; the latter choose actions by test-time search. We realize the same separation inside a single learned policy without search and ask when the hierarchy helps, by contrasting it with a flat policy trained on the same data with the same number of trainable parameters. The upper level generates a near-future latent in the space of a frozen visual encoder; the lower level generates a short action sequence toward that latent as a subgoal. On PushT, the hierarchical policy outperforms the flat policy for goals farther than those seen in training; within the training range the two are comparable, and adding far goals to training shrinks the gap at those distances. On the language-goal manipulation benchmark LIBERO-Long, the hierarchical policy does not win under any of the width allocations tested. Intervening on the future latent shows that the lower level depends on the upper level's prediction, yet on LIBERO-Long reducing prediction error does not raise closed-loop success. Together, the results suggest that a division in which only the upper level sees the goal distance, while the lower level solves a problem of fixed, short time span, favors generalization to goals beyond the training distance.

1 Introduction

Goal-conditioned visuomotor policies connect a distant final goal to the next few actions. Robot foundation models assign semantic understanding and continuous action generation to separate parts [2, 21]. Latent world models such as LeWorldModel [17], DINO-WM [29], and V-JEPA 2 [1] learn transition models in latent space, and HWM [28], Hi-LeWM [3], and H-JEPA [27] stack several time scales on top; all of these select actions by test-time search, whereas a learned policy emits actions in one forward pass. The goal-conditioned inverse-dynamics model of Nguyen et al. [20] replaces search by inference but maps the final goal directly to actions without an intermediate future. And *when* the hierarchy itself helps can only be answered by contrasting hierarchical and non-hierarchical policies with the same design, data, and number of trainable parameters.

We implement this division in a minimal form (Figure 1), echoing the cortical hierarchy of timescales, in which sensory areas integrate over short windows and higher areas over long ones [9, 19]. The upper level generates, in the latent space of a frozen encoder, the latent of the state 16 environment steps ahead; the lower level takes this predicted future latent as a subgoal and generates the next 8 actions. Only the upper level receives how far the final goal is; the lower level always solves a problem at most 16 steps ahead. Both levels are conditional flow-matching generators [14] and never evaluate or re-select candidate actions.

The closest prior work is H-JEPA [27] and its predecessor HWM [28]: H-JEPA makes both time and representation hierarchical, learns a separate latent space per level end to end, and plans by

Accepted at the UniReps 2026 Workshop (Extended Abstract Track, non-archival), Paris, December 12, 2026.

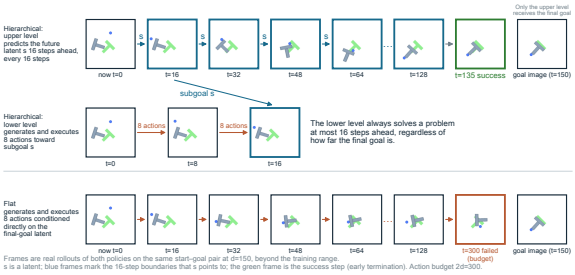


Figure 1: The hierarchical policy on a real PushT rollout at d=150, beyond the training range. The upper level predicts the future latent s 16 steps ahead every 16 steps; the lower level generates and executes 8 actions toward s, so the time span of its problem does not depend on how far the goal is. Bottom: the flat policy on the same pair, conditioned directly on the final-goal latent, fails to reach the goal within the budget $2d$=300. Blue frames at 16-step boundaries mark the displayed prediction-target times; the green frame is the success step (early termination).

gradient descent through all levels. Our hierarchy is temporal decomposition only, in the single space of a frozen encoder, and inference is a learned policy rather than action optimization; in keeping a single representation space it matches H-JEPA's control with an identity upper encoder, and H-JEPA's report that this temporal-only hierarchy outperforms flat LeWorldModel is independent evidence for a benefit of temporal decomposition. We separate time alone to tell temporal decomposition from representational hierarchy. Latent subgoals also appear in world-model and value-based hierarchies [4, 6, 8, 12, 16, 18, 22]; DreamZero [26] jointly generates future video and actions, whereas we predict the future in a frozen encoder's latent space; CW-VAE [24] models several time scales in one latent model; starting the flow from the past follows A2A [13]. We contribute a controlled comparison of a search-free hierarchical policy with a parameter-matched flat policy (Section 2), evidence that the benefit appears beyond the training goal distance and shrinks when the range is widened (Sections 3.1–3.2), and a negative result on LIBERO-Long with a substitution diagnosis (Sections 3.3–3.4). Code and pretrained weights are available at `https://github.com/miraiyochi-ai/hierarchical-latent-subgoals` and `https://huggingface.co/Miraiyochi/hierarchical-latent-subgoals`.

2 Method

Setting and notation. A policy receives camera images, proprioception, and a goal (the image at the goal time in PushT, a language instruction in LIBERO) and outputs actions, one per environment step. Supervision comes only from demonstrations, with no reward, environment interaction, or simulator state. A *latent* is the output of a frozen pretrained encoder on an image, and z is the latent of the current observation. The *ground-truth future latent* z^+ is the latent of the image actually reached in the demonstration 16 steps later (or at the goal time if earlier) and supervises the upper level. The *predicted future latent* s is what the upper level generates; it is the lower level's subgoal and the only future latent that exists at execution.

Hierarchical policy. The upper level generates s from z, conditioned on the latents of the last 16 observations, the last 16 executed actions, the goal latent g, and the remaining steps $r = \min(16, d - b)/16$, where b is the latest 16-step boundary. The lower level generates the next 8 actions from the last 8 observations, the last 8 proprioceptive states, and s, which is refreshed every 16 steps, so two chunks share one s. Both levels are flow-matching generators that start from a representation of the present rather than from noise. With velocity field v_θ and internal time $\tau \in [0, 1]$, upper inference applies six forward Euler steps from z,

$$x(0) = z, \qquad x\big(\tau + \tfrac{1}{6}\big) = x(\tau) + \tfrac{1}{6}\, v_\theta\big(x(\tau), \tau, c_H\big), \qquad s = x(1), \tag{1}$$

with $c_H = (z_{b-15:b},\, a_{b-16:b-1},\, g,\, r)$. The six iterations are internal and deterministic. The lower level generates a 512-D action latent the same way, starting from an encoding of the last 8 proprioceptive states, conditioned on an encoding of the last 8 observations and a linear projection of s, and a learned VAE decoder maps the endpoint to 8 actions (Appendix A, Figure 3).

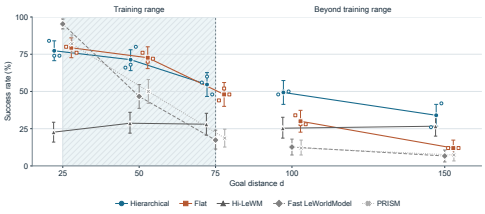


Figure 2: PushT success against goal distance. Shading: training range of our models. Large markers: means over 150 trials; bars: 95% intervals; small markers: per-seed values (protocol differences in Appendix A). Gray: public weights of Hi-LeWM [3], Fast LeWorldModel [7], and PRISM [25] on the same pairs under non-matching conditions (Appendix C).

Table 1: PushT beyond the training range (150 trials per cell, paired differences).

d	Hier.	Flat	Diff. [95% interval]
100	49.3%	30.0%	$+19.3$ [$+9.3$, $+29.3$]
150	34.0%	12.0%	$+22.0$ [$+14.0$, $+30.0$]

Table 2: LIBERO-Long parameter control (10 tasks $\times$ 50 states; widths are upper·lower).

Model	Widths	Params	Success
Flat	–·849	74.855 M	402/500 (80.4%)
Flat, lower 512	–·512	53.777 M	390/500 (78.0%)
Hierarchical	512·512	74.847 M	379/500 (75.8%)
Hier., upper 1024	1024·512	79.470 M	372/500 (74.4%)

Training and flat control. The upper loss is the flow-matching velocity error on the straight path from z to z^+ plus the six-step endpoint error to z^+; the lower loss is the analogous errors toward the encoded demonstrated actions, L1 errors of decoded actions, and a KL term (Appendix A). The lower level receives s, not z^+, during training, and the lower loss back-propagates through the upper integration. Both policies train for 30,000 updates and are evaluated with the final EMA weights without checkpoint selection. The flat policy is the same lower network receiving the final-goal latent (goal image through the same frozen encoder in PushT, language embedding in LIBERO) at the slot where s enters, with its width adjusted so that trainable parameters match within 0.1%. It receives the same 8-step observation and proprioceptive inputs as the lower level, but not the upper level's 16-step observation window, action history, or remaining-steps input.

3 Experiments

We ask: (Q1) for goals farther than seen in training, does the hierarchical policy succeed more often than the flat one? (Q2) Does the benefit survive when the training range is widened? (Q3) Does it generalize to LIBERO-Long, with language goals and two-view 7-DoF manipulation, closer to practice?

Setup. PushT [5] pushes a T-shaped block to a target pose. Using the public demonstrations of LeWorldModel [17], a task takes the state at time t and the image at $t + d$ as the goal. Training samples $d \in \{25, 50, 75\}$ uniformly; the action budget is $2d$ (Appendix A). In-range evaluation follows Hi-LeWM [3]; each distance has 150 trials (3 training seeds $\times$ 50). LIBERO-Long (`libero_10`) [15] has 10 long-procedure tasks with language instructions, encoded by frozen T5-base [23]; the upper latent space uses frozen V-JEPA 2 because the LeWorldModel-type encoder is PushT-specific. Per task, 45 demonstrations train and 5 are held out; evaluation is 10 tasks $\times$ 50 initial states, budget 520 actions (Table 3). Both policies receive the same start–goal pairs; a difference is the mean of paired per-trial differences in percentage points, with 95% bootstrap intervals stratified by training seed (10,000 resamples for PushT). The extension (Section 3.2) and LIBERO use one training seed.

3.1 PushT: the hierarchical policy wins beyond the training range (Q1)

Within the training range, d=25, 50, 75, the paired differences (hierarchical minus flat) are -2.0, -1.3, and $+6.7$ points with 95% intervals that all include 0 (Figure 2). Applying the same weights without retraining to d=100 and 150, on 150 pairs per distance from the 439 held-out demonstrations, gives Table 1: the hierarchical policy wins at both distances. Public weights of search-based methods are shown for reference (Figure 2); d=200 is in Appendix B. Substituting the upper output by the current latent z drops success to 0.7% at d=100 and 0% at d=150; if the lower level ignored s, nothing would change, so it depends on the upper output. The privileged substitution by z^+ (unavailable at execution) gives 56.7% and 39.3%, that is, $+7.3$ [-0.7, $+16.0$] and $+5.3$ [-4.7, $+15.3$] points over the predicted condition.

3.2 Widening the training range shrinks the gap (Q2)

With seed 42, retraining on $d \in \{25, 50, 75\}$ gives gaps of $+24$ (d=100) and $+26$ (d=150) points, and widening the training distances to $\{25, 50, 75, 100, 150\}$ with the same model and total updates shrinks them to $+8$ and $+4$ points (50 trials each; Figure 5, Table 4, McNemar tests in Appendix B).

3.3 LIBERO-Long: no benefit (Q3)

Because the action-generating lower level has width 512 in the hierarchical policy and 849 in the parameter-matched flat one, we add a flat policy with lower width 512 and a hierarchical policy with upper width 1024 (Table 2). The hierarchical policy is 4.6 points below the flat policy and 2.2 points below the width-matched flat policy, and widening the upper level does not help (per task: Table 5).

3.4 Diagnosis by substituting the upper output

We measure the lower level offline on the 50 held-out demonstrations (1,677 windows of lower input and next 8 demonstrated actions). In each window the upper-output slot is filled with z, s, or z^+, and we measure the action L1 error against the demonstration, averaged window→demonstration→task (Figure 6); the flat policy is reported on the same windows. For both hierarchical policies the error is largest with z, so the lower level uses the upper output, and smallest with z^+, so a future latent closer to the truth gives better actions. Widening the upper level to 1024 lowers the future-latent squared error from 0.372 to 0.366 and the action error with s from 0.0970 to 0.0965, yet closed-loop success falls from 75.8% to 74.4%. The flat policy's error, 0.0975, nearly equals the hierarchical policy's with s (0.0970), while its closed-loop success is 4.6 points higher: offline action error does not explain the closed-loop gap.

4 Discussion and limitations

When the benefit holds. In the main comparison, the paired 95% intervals exclude 0 only for models trained on $d \in \{25, 50, 75\}$ applied to d=100, 150. The hierarchical lower level always solves "8 actions toward a latent at most 16 steps ahead," whereas the flat policy's problem changes with goal distance: a goal latent beyond the training range enters its action generator directly, in a combination never seen in training, while in the hierarchical policy it reaches only the upper level. The collapse under z substitution shows that the lower level relies on this subgoal, and replacing the prediction by z^+ raises success by only $+7.3$ and $+5.3$ points (intervals include 0), so most of the $+19.3$ and $+22.0$ point gap is not accounted for by this substitution, which supports the fixed time span as a contributor; prediction accuracy and temporal decomposition are not isolated (Appendix B).

Why LIBERO-Long shows no benefit. First, training and evaluation share the 10 tasks and a language goal has no "how far ahead," so the condition under which PushT showed a benefit does not exist. Second, lower width alone does not explain the gap, and prediction error falls without closed-loop gain (Section 3.4), supporting the hypothesis that s keeps "where to go" but loses information needed for "how to move now": the lower loss back-propagates into the upper level, so s is shaped to fit both the regression to z^+ and the action reconstruction. SUNTA [11] reports that joint training of a hierarchical state-space model lets the upper conditioning erase the lower level's learning signal, and trains stage-wise instead; an upper level trained only by z^+ regression, detached from the lower loss, or a stage-wise control would test this directly. Third, the offline diagnostic averages errors evenly within and then across demonstrations and tasks, and uses demonstrator-visited states (Appendix B).

Limitations and conclusion. The main result uses 3 training seeds; the extension and LIBERO use one, and intervals resample trials only. The in-range PushT evaluation did not meet the criterion registered before training (Appendix A); the beyond-range evaluation was added afterwards on the same weights. Parameters, not compute, are matched; all evaluation is in simulation; further limitations are listed in Appendix B. Taken together, the three PushT results (the advantage beyond the training range, the parity within it, and the shrinkage as the range widens) suggest that leaving the goal distance to the upper level alone, so that the lower level's problem keeps a fixed, short time span, is a main source of this design's advantage for goals beyond the training distance. The LIBERO-Long result is consistent with this account.

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A Implementation and procedure

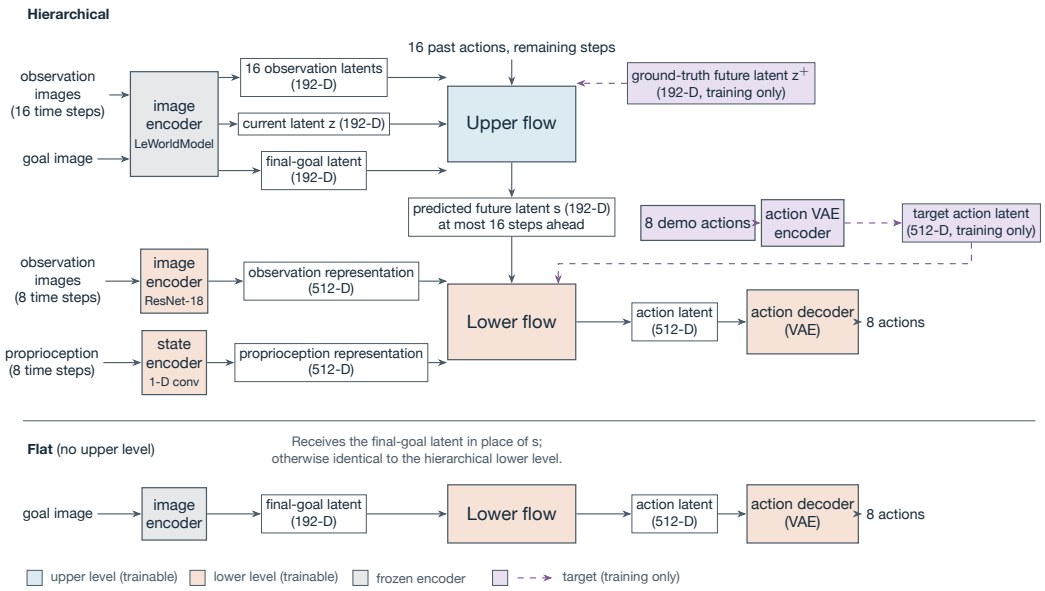


Figure 3: Hierarchical (top) and flat (bottom) policies for PushT. Starting from the frozen encoder's latent z, the upper flow generates the predicted future latent s at most 16 steps ahead, and the lower flow generates an action latent conditioned on s and decodes it into 8 actions. Training supervises the upper level with the ground-truth future latent z^+ and the lower level with demonstrated actions. The flat policy has no upper level and receives the final-goal latent in place of s.

Four time scales. Three of the four time scales live on environment steps (Figure 4). The goal distance d is the gap between start and goal time in the demonstration, $\{25, 50, 75\}$ in training and up to 200 in evaluation. The upper period 16 is the number of environment steps between upper updates and also the prediction horizon. The target time is $\min(b + 16,\ d)$ and the remaining-steps input is $\min(16,\ d - b)/16$; the last segment is truncated at the goal time, after which the last s is held. The lower chunk length 8 is the number of actions generated at once and executed in order. The six integration steps are the internal iterations of the flow from start to endpoint, during which the environment does not advance.

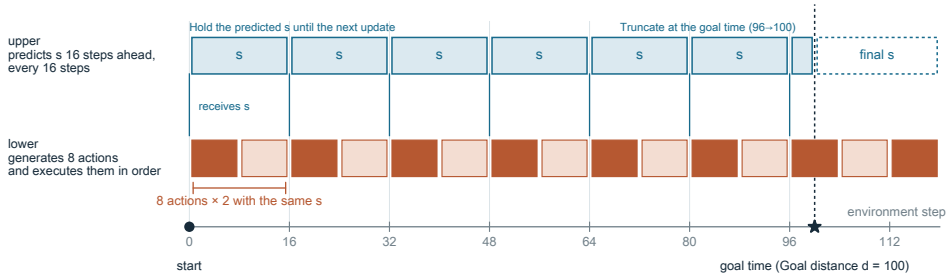


Figure 4: Time scales of the upper and lower levels (example d=100). The horizontal axis is environment steps; • is the start and ⋆ the goal time. The upper level generates s (squares) 16 steps ahead every 16 steps, truncated at the goal time and held afterwards. The lower level generates and executes 8 actions under the received s; shading distinguishes the first and second chunk under the same s.

PushT inputs and networks. The upper level uses the 192-D representation of the frozen LeWorldModel-type ViT-tiny encoder distributed with Hi-LeWM [3] and receives 16 observation

latents, 16 past actions, the goal image latent, a mask indicating whether history exists, and the normalized remaining steps. Its velocity field is a 4-layer MLP of hidden width 512; the observation window is the concatenation of 16 latents mapped to 512-D by one layer, and the action history likewise. The lower level uses a trainable ResNet-18 image encoder (GroupNorm, random initialization), a 1-D convolutional encoder over 8 proprioceptive states, and a linear projection of the upper latent. The lower image encoder in LIBERO is frozen V-JEPA 2 features with a trainable linear map for both policies. The flat policy has no upper level; the goal image encoded and normalized by the same frozen encoder enters the projection slot where s enters in the hierarchical policy. It receives no action history and no remaining steps.

Lower-level flow and losses. The 8 actions $a_{u+1:u+8}$ are encoded and decoded by a learned VAE with a 512-D latent. The lower flow starts at $y(0) = e_p(p_{u-7:u})$, an encoding of the last 8 proprioceptive states, with condition $c_L = (e_o(o_{u-7:u}),\, Ws)$, and applies the same six Euler steps with velocity field u_ϕ; the decoder D maps $y(1)$ to $\hat{a}_{u+1:u+8}$. With $\tau \sim \mathcal{U}[0,1]$ and $x(\tau) = (1-\tau)z + \tau z^+$ (on a straight path the true velocity is $z^+ - z$ for every τ), the upper loss is

$$\mathcal{L}_H = \big\| v_\theta\big(x(\tau), \tau, c_H\big) - (z^+ - z) \big\|^2 + \|\hat{s} - z^+\|^2, \tag{2}$$

where $\hat{s}$ is the six-step endpoint. The lower level encodes the demonstrated actions a with the VAE encoder into the target latent y^+; with $y_0 = e_p(p_{u-7:u})$ and the straight path from y_0 to y^+,

$$\mathcal{L}_L = \big\| u_\phi\big(y(\tau), \tau, c_L\big) - (y^+ - y_0) \big\|^2 + \|\hat{y} - y^+\|^2 + \tfrac{1}{2}\|D(\hat{y}) - a\|_1 + \tfrac{1}{2}\|D(y^+) - a\|_1 + 10^{-6}\,\mathrm{KL}. \tag{3}$$

The two losses are summed with weight 1. Each update feeds 128 upper samples and 128 lower samples. No gradient reaches the frozen encoders.

One training update of the hierarchical policy (PushT). (1) Upper sample: draw d uniformly from $\{25, 50, 75\}$ from a training demonstration, a start time t and goal time $t + d$, and a 16-step boundary b; take the 16 observation images up to $t+b$, the 16 preceding actions, the goal-time image, the remaining steps $\min(16, d-b)/16$, and the target image at $t + \min(b+16, d)$. At b=0 the action history is zero-filled and masked out. Images are encoded by the frozen encoder and standardized per component with the mean and standard deviation over all frames of all training demonstrations. (2) Lower sample: independently draw a time u on an 8-step grid, the 8 observation images and proprioceptive states up to $t + u$, the following 8 actions, and the full upper input at the preceding boundary $16\lfloor u/16 \rfloor$. (3) Upper loss: velocity and endpoint errors. (4) Lower loss: integrate the same upper network six times from the lower sample's upper input to obtain s, not detached, so the gradient flows to the upper level; compute velocity and endpoint errors, decoded-action L1 errors (generated and target sides, 0.5 each), and the KL term. (5) Update: sum the two losses, step AdamW over all trainable parameters, and update the EMA weights. The flat policy skips (1) and (3) and in (4) replaces s by the standardized goal latent. LIBERO differs in input types, goal-distance handling, and normalization, but the control flow is the same.

Optimization. AdamW, learning rate 10^{-4}, $\beta = (0.95, 0.999)$, $\epsilon = 10^{-8}$, weight decay 10^{-6}, 500 warm-up updates then cosine decay, gradient clipping at 1. Effective batch 128 with micro-batch 32; 30,000 updates; the final EMA weights are used without selection.

One evaluation trial (PushT). Take a task with start time t and distance d from an evaluation demonstration and restore the physical state at t in the simulator. Restore the state at $t + d$ separately, render it as the goal image, encode and standardize it; this goal latent is the only goal information the policy receives. Initialize the observation history as 16 copies of the initial image and the proprioceptive history as 8 copies of the initial state; the action history is masked at boundary 0. At boundaries $b = 0, 16, 32, \ldots$ ($b < d$) the upper level updates s by six integration steps; the lower level generates 8 actions from the last 8 observations and states and s and sends them in order. After the goal time the upper level is not updated and the last s is kept. The trial ends at the first step at which the L2 distance of the 4-D vector of end-effector and block positions to the goal is below 20 pixels and the block angle error is below $\pi/9$; it fails when the action count reaches $2d$. In LIBERO, success is the simulator's task-completion check. Beyond-range pairs (d=100, 150) are drawn from the 439 held-out demonstrations whose goal time falls inside the demonstration, 150 per distance, whereas in-range pairs follow the Hi-LeWM protocol. McNemar tests use only the discordant trials.

Evaluation history. The criterion for the PushT main judgment was registered before training on 2026-08-10: at both d=50 and d=75, the hierarchical policy must exceed the flat policy by at least 3 points (d=50) and 5 points (d=75), with the lower bound of the 95% paired interval above 0 and a positive difference for all 3 training seeds; the criterion also covered comparisons against the z-substitution and z^+ conditions. The outcome was -1.3 at d=50 and $+6.7$ at d=75 with intervals including 0 (Section 3.1, Appendix B), so the criterion was not met. The beyond-range evaluation at d=100, 150, 200 was registered separately after this judgment and run on the same weights.

Table 3: Changes from PushT to LIBERO-Long. The 16-step upper period, 8-action chunk, six integration steps, losses, optimizer, and 30,000 updates are unchanged.

Item	PushT	LIBERO-Long
Goal	image at the goal time	frozen T5-base [23] embedding (mean of valid tokens, 768-D)
Upper latent space	frozen LeWorldModel-type encoder (ViT-tiny, distributed with Hi-LeWM), 192-D	frozen V-JEPA 2 ViT-L [1], 2 views × 1024-D concatenated
Lower image processing	trainable ResNet-18	frozen V-JEPA 2 features + trainable linear map (both policies)
Proprioception / action	2-D / 2-D	9-D / 7-D
Remaining-steps input	steps to goal time (capped at 16)	constant (demonstration length not revealed)
Evaluation	50 tasks × 3 seeds per distance, budget $2d$	10 tasks × 50 initial states, budget 520 actions

V-JEPA 2 is a video encoder; here a single image is duplicated into two frames, spatially averaged, and normalized without learned parameters. The upper target is the latent of the observation 16 steps after the boundary (or the last frame if the demonstration ends earlier). The budget is 520 actions with 10 initial settling steps. LIBERO-Long data are the fixed public release; demonstrations 0–44 of each task are used for training and 45–49 for diagnostics. Diagnostic windows advance the lower anchor on an 8-step grid, with boundary $16\lfloor \text{anchor}/16 \rfloor$, lower observations from anchor-7 to anchor, target actions from anchor$+1$ to anchor$+8$, and the upper target at row $\min(\text{boundary} + 16,\ \text{length} - 1)$. A window bundles the lower input (the last 8 observations and states) and the next 8 target actions together with the upper input and target latent defined at the boundary the window belongs to.

B Detailed results

PushT within the training range. Over 3 training seeds × 50 trials, the paired differences (hierarchical minus flat) with 95% intervals are -2.0 [-7.3, $+3.3$] at d=25, -1.3 [-8.0, $+5.3$] at d=50, and $+6.7$ [-1.3, $+14.7$] at d=75.

PushT beyond the range, per seed. Hierarchical minus flat is $+14$, $+22$, $+22$ points for seeds 42, 43, 44 at d=100 and $+14$, $+22$, $+30$ at d=150 (the success counts underlying Table 1: 74/150 vs. 45/150 and 51/150 vs. 18/150). The z-substitution condition succeeds in 0.7% at d=100, 0% at d=150, and 0% at d=200.

Table 4: Extension experiment of Section 3.2 (training and evaluation seed 42, 50 trials per distance), trained separately from the three main seeds. Five-distance training allots 768,000 lower samples per distance, a change in allocation under the fixed total. For the retrained $\{25, 50, 75\}$ model, the pooled d=25, 50, 75 counts are hierarchical 98/150 and flat 99/150.

d	Trained on 25–75: flat / hier.	Trained on 25–150: flat / hier.
25	82% / 76%	72% / 78%
50	68% / 62%	56% / 60%
75	48% / 58%	30% / 48%
100	36% / 60%	44% / 52%
150	10% / 36%	38% / 42%

Substitution by the ground-truth future latent in PushT. With the same 3 seeds and 150 trials, the privileged closed-loop condition reaches 56.7% at d=100 and 39.3% at d=150, that is, $+7.3$ [-0.7, $+16.0$] and $+5.3$ [-4.7, $+15.3$] points over the predicted condition. At d=200 the hierarchical policy reaches 18.0% and the flat policy 6.0% (difference $+12.0$ [$+5.3$, $+18.7$]); the privileged

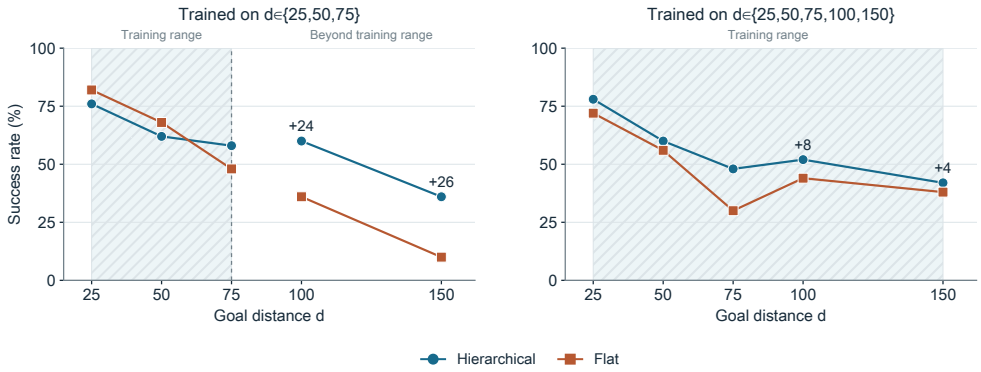


Figure 5: PushT success when the training range is widened (50 trials each, seed 42). Left: trained on $d \in \{25, 50, 75\}$; right: trained on $d \in \{25, 50, 75, 100, 150\}$; shading marks each training range. Numbers above d=100, 150 are the hierarchical minus flat differences in points.

condition reaches 31.3%, +13.3 [+6.0, +20.7] over the predicted condition. Because the substitution uses futures from fixed demonstrations, it is not a ceiling for perfect prediction at policy-visited states. Only 26 held-out demonstrations are longer than 200 steps, so the same demonstrations are reused often, and d=200 is treated as a diagnostic measurement rather than a hypothesis test.

McNemar tests for the extension. In the five-distance training, the trials in which only the hierarchical or only the flat policy succeeded number 7 vs. 3 at d=100 and 6 vs. 4 at d=150 (exact McNemar $p = 0.344$ and 0.754); this tests hierarchical vs. flat within that training, not the difference of differences. Under the fixed total of updates, five distances also change the samples allotted per distance.

Training time. In five-distance training on the same H100, 30,000 updates took 15.5 hours for the hierarchical policy and 10.6 hours for the flat one; this is not a controlled inference-speed measurement.

Another LIBERO transfer experiment. Transferring the weights of Section 3.3 (trained on `libero_10`) to LoHo [10] (5 tasks × 50 initial states, budget 1,040) without retraining gives 0/250 for both policies. In a separate run (seed 42) trained on the 129 files and 6,425 demonstrations of LIBERO-Logic without logic annotations for 60,000 updates per policy, LIBERO-Long gives hierarchical 293/500 (58.6%) vs. flat 314/500 (62.8%), and LoHo gives 2/250 (0.8%) vs. 0/250; uniform task sampling also changes the per-task training opportunity.

Table 5: LIBERO-Long per task (`libero_10` order), 50 trials each.

Task	Flat	Flat (512)	Hier.	Hier. (1024)
0	96%	96%	98%	100%
1	92%	96%	88%	74%
2	88%	88%	88%	98%
3	88%	78%	92%	92%
4	82%	78%	76%	80%
5	42%	62%	30%	12%
6	78%	68%	64%	68%
7	72%	68%	60%	66%
8	80%	60%	86%	66%
9	86%	86%	76%	88%

Table 6: Paired success breakdown on the same tasks and initial states (single training seed).

Comparison (left − right)	Diff. (pt)	Left only / right only
Hier. − Flat	−4.6	50 / 73
Flat (512) − Flat	−2.4	55 / 67
Hier. (1024) − Hier.	−1.4	55 / 62
Hier. − Flat (512)	−2.2	68 / 79
Hier. (1024) − Flat	−6.0	49 / 79

LIBERO diagnosis details. The diagnostic averages errors within demonstrations, then across demonstrations and tasks, with no extra weight on decisive moments such as grasping or re-placing, and its inputs are demonstrator-visited states, whose distribution differs from the states a policy visits in closed loop. Using the current latent z itself as the future estimate gives a squared error of

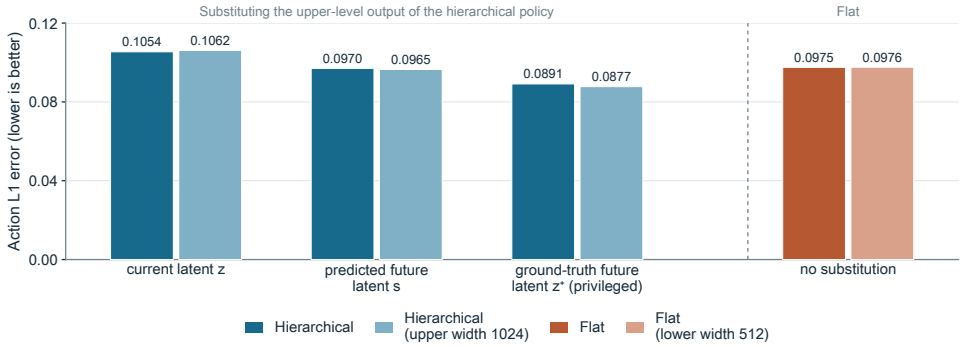


Figure 6: Substituting the upper output in LIBERO-Long and the resulting action L1 error (50 held-out demonstrations, 1,677 windows, task-balanced mean). The left three groups give the hierarchical lower level z, s, or z^+; the right two bars are the flat policies without substitution.

0.560 against z^+, so the upper level moves z toward z^+. In PushT both policies receive the goal in the same frozen latent space, whereas in LIBERO the flat policy receives the language embedding directly and the hierarchical lower level receives goal information only through the generated latent; this difference in information paths is a test point for the hypothesis of Section 4. Paired success breakdowns on the same tasks and initial states are in Table 6.

Further limitations. The hierarchical-minus-flat difference includes, besides the presence of a hierarchy, the differences in the observation window, action history, and remaining-steps inputs, and the substitution experiment removes future information and changes the lower level's input distribution in one intervention. The prediction horizon, update period, and chunk length are fixed at 16, 16, and 8; we do not compare alternative settings. From PushT to LIBERO the goal type, the lower visual representation, and the observation and action dimensions change together. The frozen encoders are external pretrained weights; we have not checked whether the held-out demonstrations were included in their pretraining data. Inference speed was not measured under controlled conditions.

C Reference evaluation of public weights and published values

Re-evaluation of public weights. We evaluated the public weights of Fast LeWorldModel [7], PRISM [25], Hi-LeWM [3], and DINO-WM [29] on the same start–goal pairs as Section 3.1 (150 per distance, budget $2d$; Table 7). Intervals are 95% bootstrap intervals stratified by the task-sampling seed, not a comparison of three independently trained models. Hi-LeWM uses the original paper's d=75 constrained setting (upper horizon 2, replanning interval 5, lower horizon 2) at every distance; its values are below the paper's per-distance tuned values (Table 8).

Table 7: Success rate (%) of re-evaluated public weights (Hi-LeWM [3], Fast LeWorldModel [7], PRISM [25], DINO-WM [29]) with 95% intervals, on the pairs of Section 3.1. DINO-WM was evaluated at d=100, 150 only.

Method	d=25	d=50	d=75	d=100	d=150
Hi-LeWM	22.7 [16.0, 29.3]	28.7 [22.0, 36.0]	28.0 [20.7, 35.3]	25.3 [18.7, 32.7]	26.7 [20.0, 34.0]
Fast LeWorldModel	95.3 [92.0, 98.7]	46.7 [38.7, 54.7]	17.3 [11.3, 24.0]	12.7 [7.3, 18.0]	6.7 [2.7, 10.7]
PRISM	82.0 [76.0, 88.0]	50.0 [42.0, 58.0]	18.7 [12.7, 24.7]	12.0 [7.3, 17.3]	7.3 [3.3, 12.0]
DINO-WM	–	–	–	11.3	4.7

These methods differ from ours in the image-generation path, training range, and control and stopping conditions; the initial and goal images of Fast LeWorldModel, Hi-LeWM, and PRISM come from the dataset, whereas Section 3.1 re-renders them in the environment. DINO-WM also differs in environment and success check and is therefore not drawn in Figure 2; its shape-coverage values from a separate computation are 1.33% and 0.00%, distinct from the official check. Fast LeWorldModel was evaluated with the implementation as of 2026-09-13; the later paper v2 was not evaluated.

Published values. Table 8 lists, for papers that report success by goal distance under the same definition as ours (times t and $t + d$ of the same demonstration), the value of the best setting at each distance. Fast LeWorldModel, PRISM, and DINO-WM do not report success by goal distance. No paper could be confirmed to match ours in action budget, success check, and goal-image generation. We have not retrained SAGE [4], HIQL [22], FF-JEPA [18], hierarchical flow Q-learning [6], Director [8], Play-LMP [16], the goal-conditioned inverse-dynamics model [20], or H-JEPA [27] under matching conditions.

Table 8: Published PushT success by goal distance (best per-distance setting in each paper). d and budgets are environment steps.

Paper / series	Table	Row setting	d	Success	Trials	Budget
Hi-LeWM [3]	4	Hi-LeWM, $(h_h, h_{\text{replan}}, h_l, r_l)=(2, 5, 2, 1)$, staged	25	90.7%	50×3 seeds	50
Hi-LeWM [3]	4	Hi-LeWM-C, (2,5,2,1), staged	50	64.0%	50×3 seeds	100
Hi-LeWM [3]	4	Hi-LeWM-C, (2,5,2,1)	75	32.7%	50×3 seeds	150
LeWorldModel flat planning [3]	3	LeWorldModel, h=5, r=5	25	94%	50×3 seeds	50
LeWorldModel flat planning [3]	3	LeWorldModel, h=5, r=5	50	52.7%	50×3 seeds	100
LeWorldModel flat planning [3]	3	LeWorldModel, h=5, r=5	75	18%	50×3 seeds	150
HWM [28]	2	DINO-WM (hier.)	25	89%	n/a	n/a
HWM [28]	2	DINO-WM (hier.)	50	78%	n/a	n/a
HWM [28]	2	DINO-WM (hier.)	75	61%	n/a	n/a
H-JEPA [27]	9	Push-T, 2 levels, H-JEPA column (best planning depth)	75	45.3%	50×3 seeds	90